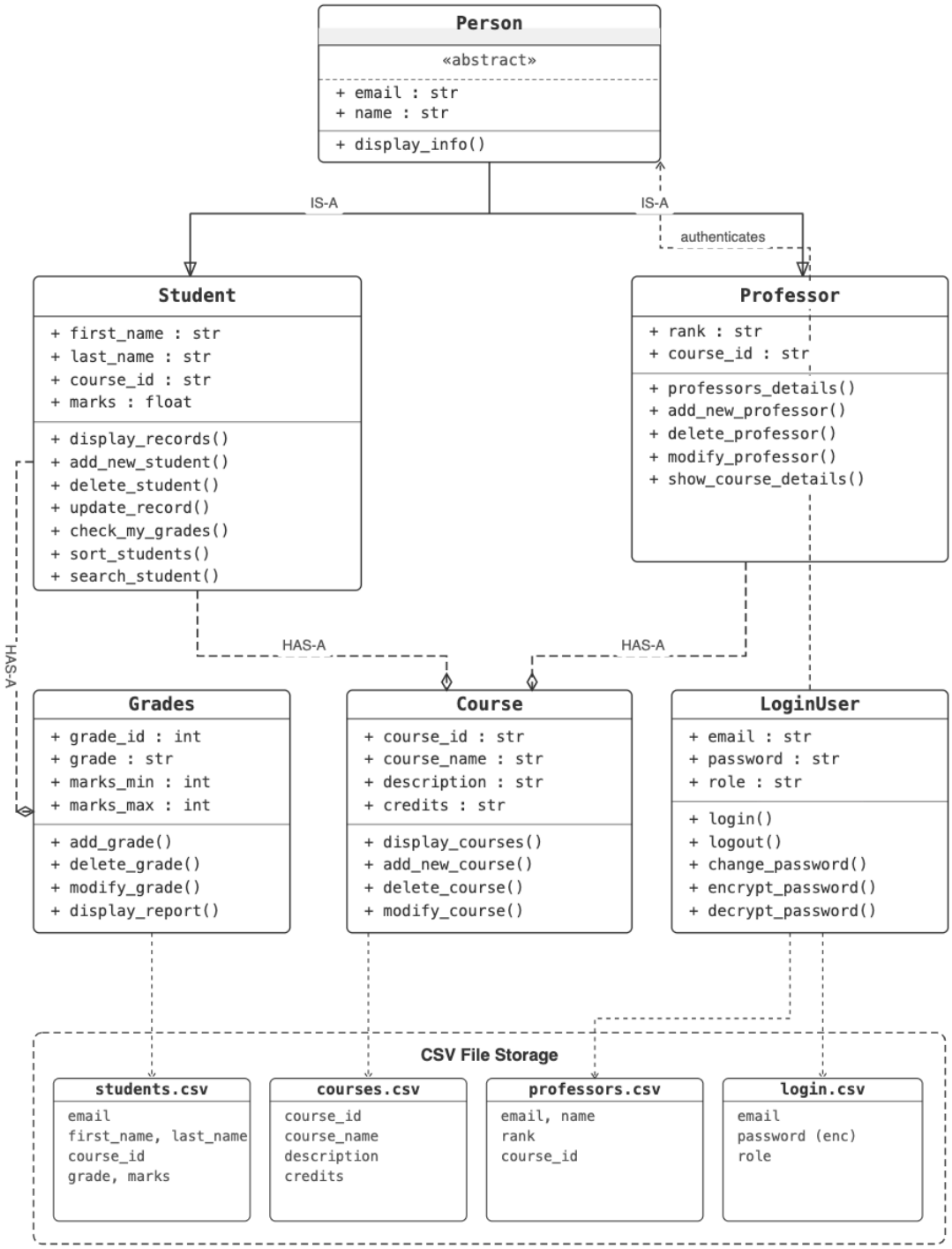
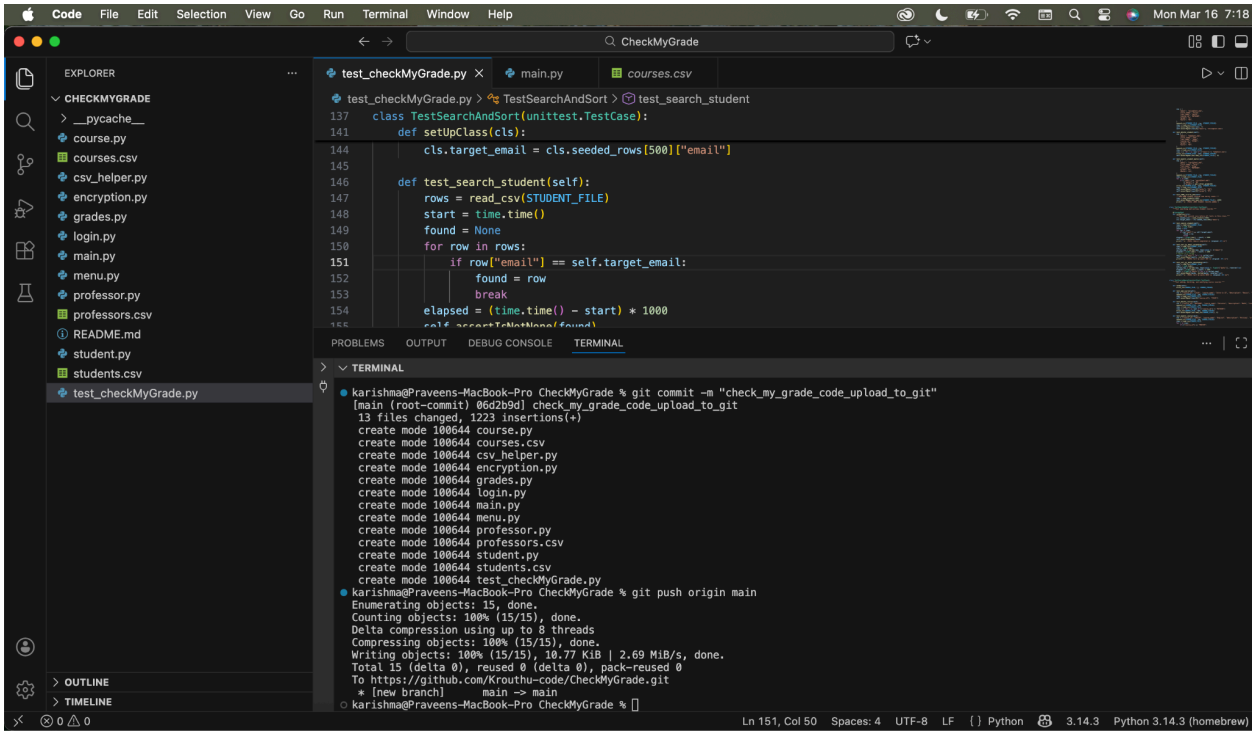
Object Oriented Design (OOD) for CheckMyGrade Application

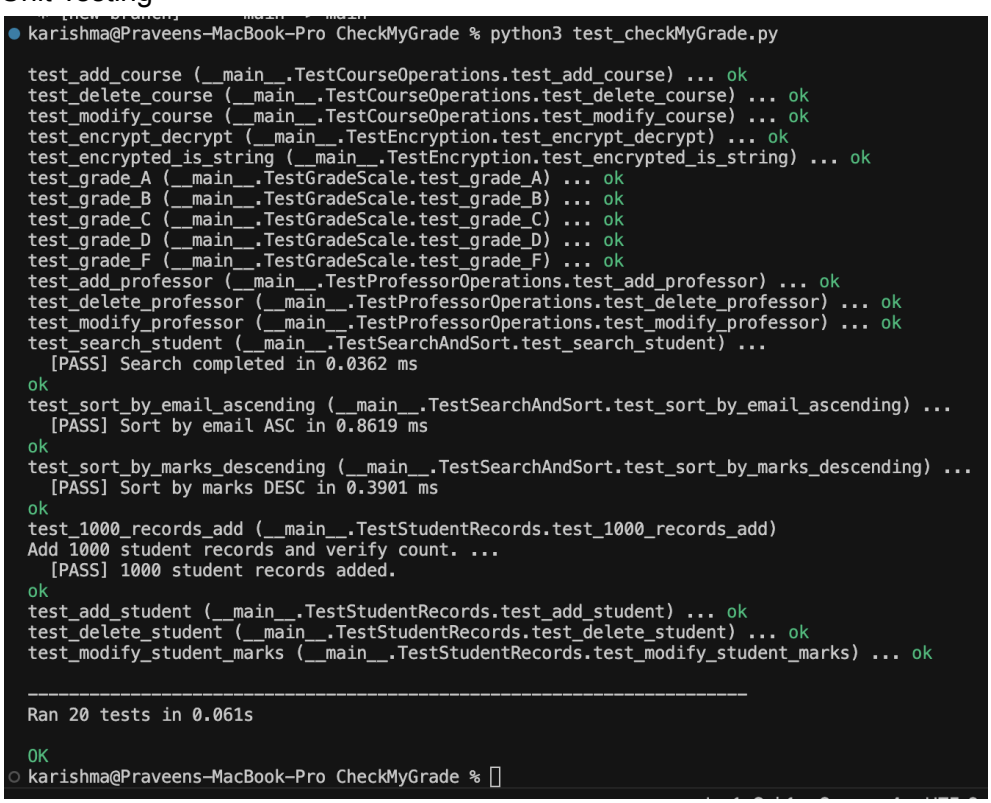


Github Screenshots:




Github Link : [https://github.com/Krouthu-code/CheckMyGrade](https://github.com/Krouthu-code/CheckMyGrade)

Unit Testing



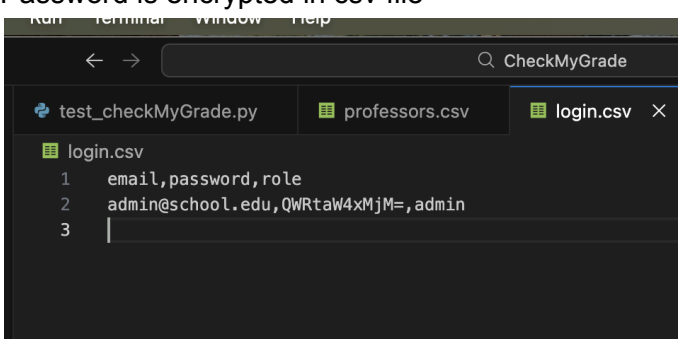
```
karishma@Praveens-MacBook-Pro CheckMyGrade % python3 test_checkMyGrade.py

test_add_course (__main__.TestCourseOperations.test_add_course) ... ok
test_delete_course (__main__.TestCourseOperations.test_delete_course) ... ok
test_modify_course (__main__.TestCourseOperations.test_modify_course) ... ok
test_encrypt_decrypt (__main__.TestEncryption.test_encrypt_decrypt) ... ok
test_encrypted_is_string (__main__.TestEncryption.test_encrypted_is_string) ... ok
test_grade_A (__main__.TestGradeScale.test_grade_A) ... ok
test_grade_B (__main__.TestGradeScale.test_grade_B) ... ok
test_grade_C (__main__.TestGradeScale.test_grade_C) ... ok
test_grade_D (__main__.TestGradeScale.test_grade_D) ... ok
test_grade_F (__main__.TestGradeScale.test_grade_F) ... ok
test_add_professor (__main__.TestProfessorOperations.test_add_professor) ... ok
test_delete_professor (__main__.TestProfessorOperations.test_delete_professor) ... ok
test_modify_professor (__main__.TestProfessorOperations.test_modify_professor) ... ok
test_search_student (__main__.TestSearchAndSort.test_search_student) ...
  [PASS] Search completed in 0.0362 ms
ok
test_sort_by_email_ascending (__main__.TestSearchAndSort.test_sort_by_email_ascending) ...
  [PASS] Sort by email ASC in 0.8619 ms
ok
test_sort_by_marks_descending (__main__.TestSearchAndSort.test_sort_by_marks_descending) ...
  [PASS] Sort by marks DESC in 0.3901 ms
ok
test_1000_records_add (__main__.TestStudentRecords.test_1000_records_add)
Add 1000 student records and verify count. ...
  [PASS] 1000 student records added.
ok
test_add_student (__main__.TestStudentRecords.test_add_student) ... ok
test_delete_student (__main__.TestStudentRecords.test_delete_student) ... ok
test_modify_student_marks (__main__.TestStudentRecords.test_modify_student_marks) ... ok

----------------------------------------------------------------------
Ran 20 tests in 0.061s

OK
karishma@Praveens-MacBook-Pro CheckMyGrade %
```

Output Screenshots:

Register and Login

```
karishma@Praveens-MacBook-Pro CheckMyGrade % python3 main.py
==================================================
     Welcome to CheckMyGrade Application
==================================================

1. Login
2. Register
3. Exit
Enter choice: 2
Enter email: admin@school.edu
Enter password: Admin123
Enter role (student/professor/admin): admin
Registered successfully as admin!

1. Login
2. Register
3. Exit
Enter choice:   1
Enter email: admin@school.edu
Enter password: Admin123

Login successful! Welcome, admin@school.edu (admin)

==================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
==================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice:
```

Password is encrypted in csv file




Viewing Student records as admin

```
--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 1

--- Student Management ---
1. View all students
2. Add student
3. Delete student
4. Update student marks
5. Search student
6. Sort students
7. Course statistics
Enter choice: 1

____________________________________________________________


Record #1
  Email     : carol@test.edu
  Name      : Carol King
  Course    : MATH300
  Grade     : B
  Marks     : 88
____________________________________________________________
```

Checking courses

```
--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 2

--- Course Management ---
1. View all courses
2. Add course
3. Delete course
4. Modify course
Enter choice: 1

_____________________________________________________


Course #1
  Course ID   : ENG150
  Name        : Advanced English
  Description : Writing
  Credits     : 3
_____________________________________________________
```

Checking professors

```
--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 3

--- Professor Management ---
1. View all professors
2. Add professor
3. Delete professor
4. Modify professor
Enter choice: 1

____________________________________________________


Professor #1
  Email     : prof3@test.edu
  Name      : Dr. Brown
  Rank      : Senior Professor
  Course ID : MATH300
____________________________________________________
```

Grade Report by student email

```
--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 4

--- Grade Reports ---
1. Report by course
2. Report by student
3. Report by professor
Enter choice: 2
Enter student email: carol@test.edu

Grade Report for Student: carol@test.edu
____________________________________________________
  Course: MATH300  |  Grade: B  |  Marks: 88
```

Grade Report By course id

```
--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 4

--- Grade Reports ---
1. Report by course
2. Report by student
3. Report by professor
Enter choice: 1
Enter course ID: MATH300

Grade Report for Course: MATH300
____________________________________________________
  carol@test.edu  |  Carol King  |  Grade: B  |  Marks: 88
```

Grade Scale

```
--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 5

Grade Scale:
_____________________________
  Grade: A  |  Range: 90 - 100
  Grade: B  |  Range: 80 - 89
  Grade: C  |  Range: 70 - 79
  Grade: D  |  Range: 60 - 69
  Grade: F  |  Range: 0 - 59
```

Student Management Screenshots

Add student

```
--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 1

--- Student Management ---
1. View all students
2. Add student
3. Delete student
4. Update student marks
5. Search student
6. Sort students
7. Course statistics
Enter choice: 2
Student email: zara@mycsu.edu
First name: Zara
Last name: Ahmed
Course ID: DATA200
Marks (0-100): 89
Student Zara Ahmed added successfully!
```

View all students

```
--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 1

--- Student Management ---
1. View all students
2. Add student
3. Delete student
4. Update student marks
5. Search student
6. Sort students
7. Course statistics
Enter choice: 1

────────────────────────────────────────────────────────────


Record #1
  Email     : carol@test.edu
  Name      : Carol King
  Course    : MATH300
  Grade     : B
  Marks     : 88

Record #2
  Email     : zara@mycsu.edu
  Name      : Zara Ahmed
  Course    : DATA200
  Grade     : B
  Marks     : 89
────────────────────────────────────────────────────────────
```

Update Student Record

```
Record #1
  Email     : carol@test.edu
  Name      : Carol King
  Course    : MATH300
  Grade     : B
  Marks     : 88

Record #2
  Email     : zara@mycsu.edu
  Name      : Zara Ahmed
  Course    : DATA200
  Grade     : B
  Marks     : 89
--------------------------------------------------------------------


====================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
====================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 1

--- Student Management ---
1. View all students
2. Add student
3. Delete student
4. Update student marks
5. Search student
6. Sort students
7. Course statistics
Enter choice: 4
Enter student email to update: zara@mycsu.edu
Current marks: 89
Enter new marks (0-100): 81
Student record updated.
```

View updated Records

```
====================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
====================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 1

--- Student Management ---
1. View all students
2. Add student
3. Delete student
4. Update student marks
5. Search student
6. Sort students
7. Course statistics
Enter choice: 1

____________________________________________________________

Record #1
  Email     : carol@test.edu
  Name      : Carol King
  Course    : MATH300
  Grade     : B
  Marks     : 88

Record #2
  Email     : zara@mycsu.edu
  Name      : Zara Ahmed
  Course    : DATA200
  Grade     : B
  Marks     : 81
____________________________________________________________
```

Search Student

```
==================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
==================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 1

--- Student Management ---
1. View all students
2. Add student
3. Delete student
4. Update student marks
5. Search student
6. Sort students
7. Course statistics
Enter choice: 5
Enter student email to search: zara@mycsu.edu

Student found:
  Email     : zara@mycsu.edu
  Name      : Zara Ahmed
  Course    : DATA200
  Grade     : B
  Marks     : 81
Search time: 1.1091 ms
```

Sort students

```
==================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
==================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 1

--- Student Management ---
1. View all students
2. Add student
3. Delete student
4. Update student marks
5. Search student
6. Sort students
7. Course statistics
Enter choice: 6

Sort by:
1. Email (ascending)
2. Email (descending)
3. Marks (ascending)
4. Marks (descending)
Enter choice: 3

Sorted results (3.5851 ms):
------------------------------------------------------------------
  1. zara@mycsu.edu  |  Zara Ahmed  |  Marks: 81  |  Grade: B
  2. carol@test.edu  |  Carol King  |  Marks: 88  |  Grade: B
```

Sort by email descending

```
  CheckMyGrade | Logged in as: admin@school.edu (admin)
=====================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 1

--- Student Management ---
1. View all students
2. Add student
3. Delete student
4. Update student marks
5. Search student
6. Sort students
7. Course statistics
Enter choice: 6

Sort by:
1. Email (ascending)
2. Email (descending)
3. Marks (ascending)
4. Marks (descending)
Enter choice: 2

Sorted results (0.0620 ms):
------------------------------------------------------------------------
  1. zara@mycsu.edu  |  Zara Ahmed  |  Marks: 81  |  Grade: B
  2. carol@test.edu  |  Carol King  |  Marks: 88  |  Grade: B

-----------------------------------------------------
```

Course statistics

```
====================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
====================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 1

--- Student Management ---
1. View all students
2. Add student
3. Delete student
4. Update student marks
5. Search student
6. Sort students
7. Course statistics
Enter choice: 7
Enter course ID: DATA200

Course: DATA200
  Number of students : 1
  Average marks      : 81.00
  Median marks       : 81.00
```

Delete student

```
==================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 1

--- Student Management ---
1. View all students
2. Add student
3. Delete student
4. Update student marks
5. Search student
6. Sort students
7. Course statistics
Enter choice: 3
Enter student email to delete: zara@mycsu.edu
Student deleted successfully.
```

```
=====================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
=====================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 1

--- Student Management ---
1. View all students
2. Add student
3. Delete student
4. Update student marks
5. Search student
6. Sort students
7. Course statistics
Enter choice: 1

____________________________________________________________


Record #1
  Email     : carol@test.edu
  Name      : Carol King
  Course    : MATH300
  Grade     : B
  Marks     : 88
____________________________________________________________
```

Course Management
View all courses

```
==================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
==================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 2

--- Course Management ---
1. View all courses
2. Add course
3. Delete course
4. Modify course
Enter choice: 1

__________________________________________________


Course #1
  Course ID   : ENG150
  Name        : Advanced English
  Description : Writing
  Credits     : 3
__________________________________________________
```

Add course

```
====================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
====================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 2

--- Course Management ---
1. View all courses
2. Add course
3. Delete course
4. Modify course
Enter choice: 2
Course ID: PHY101
Course name: Physics I
Description: Mechanics and Waves
Credits: 4
Course 'Physics I' added successfully!
```

```
karishma@Praveens-MacBook-Pro CheckMyGrade % python3 main.py

====================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
====================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 2

--- Course Management ---
1. View all courses
2. Add course
3. Delete course
4. Modify course
Enter choice: 1

____________________________________________________

Course #1
  Course ID   : ENG150
  Name        : Advanced English
  Description : Writing
  Credits     : 3

Course #2
  Course ID   : PHY101
  Name        : Physics I
  Description : Mechanics and Waves
  Credits     : 4
____________________________________________________
```

Modify course

```
====================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
====================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 2

--- Course Management ---
1. View all courses
2. Add course
3. Delete course
4. Modify course
Enter choice: 4
Enter course ID to modify: PHY101
Current name: Physics I
New name (press Enter to keep): Physics I — Advanced
Current description: Mechanics and Waves
New description (press Enter to keep):
Course updated.

====================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
====================================================
```

```
--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 2

--- Course Management ---
1. View all courses
2. Add course
3. Delete course
4. Modify course
Enter choice: 1

____________________________________________________


Course #1
  Course ID   : ENG150
  Name        : Advanced English
  Description : Writing
  Credits     : 3

Course #2
  Course ID   : PHY101
  Name        : Physics I – Advanced
  Description : Mechanics and Waves
  Credits     : 4
____________________________________________________
```

Delete Course

```
=====================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
=====================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 2

--- Course Management ---
1. View all courses
2. Add course
3. Delete course
4. Modify course
Enter choice: 3
Enter course ID to delete: PHY101
Course deleted.
```

```
=====================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
=====================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 2

--- Course Management ---
1. View all courses
2. Add course
3. Delete course
4. Modify course
Enter choice: 1

-----------------------------------------------------

Course #1
  Course ID   : ENG150
  Name        : Advanced English
  Description : Writing
  Credits     : 3
-----------------------------------------------------
```

Professor Management
Add Professor

```
====================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
====================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 3

--- Professor Management ---
1. View all professors
2. Add professor
3. Delete professor
4. Modify professor
Enter choice: 2
Professor email: james@school.edu
Professor name: James Carter
Rank (e.g. Junior/Senior Professor): Junior Professor
Course ID: BIO101
Professor 'James Carter' added successfully!
```

List all professors

```
--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 3

--- Professor Management ---
1. View all professors
2. Add professor
3. Delete professor
4. Modify professor
Enter choice: 1

____________________________________________________


Professor #1
  Email     : prof3@test.edu
  Name      : Dr. Brown
  Rank      : Senior Professor
  Course ID : MATH300

Professor #2
  Email     : james@school.edu
  Name      : James Carter
  Rank      : Junior Professor
  Course ID : BIO101
____________________________________________________
```

Modify Professor

```
====================================================
  CheckMyGrade | Logged in as: admin@school.edu (admin)
====================================================

--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 3

--- Professor Management ---
1. View all professors
2. Add professor
3. Delete professor
4. Modify professor
Enter choice: 4
Enter professor email to modify: james@school.edu
Current name: James Carter
New name (press Enter to keep):
Current rank: Junior Professor
New rank (press Enter to keep): Senior Professor
Current course ID: BIO101
New course ID (press Enter to keep):
Professor record updated.
```

```
--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 3

--- Professor Management ---
1. View all professors
2. Add professor
3. Delete professor
4. Modify professor
Enter choice: 1

____________________________________________________


Professor #1
  Email     : prof3@test.edu
  Name      : Dr. Brown
  Rank      : Senior Professor
  Course ID : MATH300

Professor #2
  Email     : james@school.edu
  Name      : James Carter
  Rank      : Senior Professor
  Course ID : BIO101
____________________________________________________
```

Delete Professor

```
--- ADMIN MENU ---
1.  Student Management
2.  Course Management
3.  Professor Management
4.  Grade Reports
5.  Grade Scale
6.  Change password
7.  Logout
Enter choice: 3

--- Professor Management ---
1. View all professors
2. Add professor
3. Delete professor
4. Modify professor
Enter choice: 3
Enter professor email to delete: james@school.edu
Professor deleted.
```